

Small Intestine - Anatomy Notes

SMALL INTESTINE

Regular Exam Notes - Duodenum, Jejunum and Ileum

Designed in the same filled-page style as the latest gallbladder notes: larger diagrams, explanatory image text, compact but readable boxes, and 11 pt body font.

At a Glance

- Extent: from pylorus to ileocaecal junction.
- Length: about 6 m in the adult.
- Parts: duodenum, jejunum and ileum.
- Main function: digestion and absorption of nutrients.

Exam Focus

- Duodenum: parts, papillae, ligament of Treitz, ulcer site.
- Jejunum vs ileum differences: wall, lumen, folds, Peyer patches, arcades.
- Mesentery: root, surface marking, clinical significance.
- Blood, lymph and nerve supply with clinical correlations.

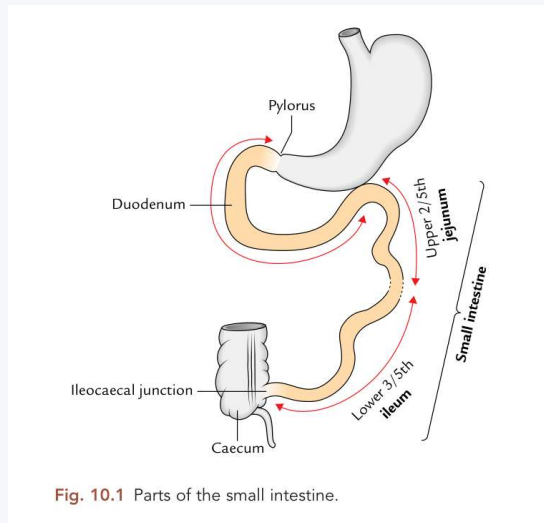


Fig. 10.1 Parts of the small intestine.

Fig. 10.1 - Parts of small intestine.

How to read this diagram

The diagram shows that the duodenum is the proximal C-shaped part, while the mobile part is divided into upper two-fifth jejunum and lower three-fifth ileum.

There is no sharp external line between jejunum and ileum. They are recognized by wall thickness, vascular pattern, folds and lymphoid tissue.

Clinically, the terminal ileum is important because it opens into the caecum at the ileocaecal junction and is related to Meckel diverticulum and ileocaecal diseases.

DEFINITION AND GENERAL FEATURES

Definition

The small intestine is the longest part of the alimentary canal. It begins at the pylorus and ends at the ileocaecal junction. It is divided into duodenum, jejunum and ileum. The duodenum is mostly fixed and retroperitoneal, whereas the jejunum and ileum are mobile and suspended by the mesentery.

Functional importance

- Duodenum receives acidic chyme from stomach and mixes it with bile and pancreatic enzymes.
- Jejunum and ileum complete digestion and

Peritoneal position

- First 2.5 cm of first part of duodenum is intraperitoneal.
- Rest of duodenum is mostly secondarily

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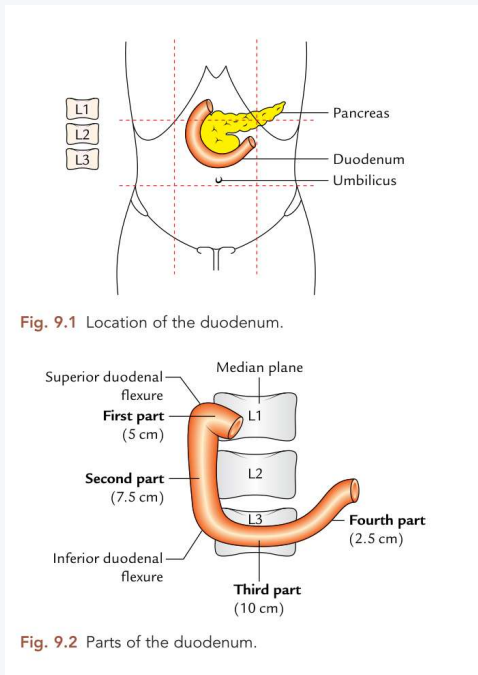
absorb most nutrients.

- Large mucosal surface area is produced by circular folds, villi and microvilli.
- Peyer patches in ileum are important for gut immunity.

retroperitoneal.

- Jejunum and ileum are intraperitoneal and mobile.
- Mobility is due to the mesentery of the small intestine.

DUODENUM



Figs. 9.1 and 9.2 - Location and parts of duodenum.

Key anatomical points

The duodenum is the first, shortest, widest and most fixed part of the small intestine. It is about 25 cm long and extends from the pylorus to the duodenojejunal flexure.

It is C-shaped and its concavity encloses the head of pancreas. It lies mainly above the umbilicus opposite L1, L2 and L3 vertebrae.

Its four parts are: superior part 5 cm, descending part 7.5 cm, horizontal part 10 cm and ascending part 2.5 cm.

Parts and main relations of the duodenum

- First part: begins at pylorus, passes upward/backward/right; related anteriorly to quadrate lobe of liver and gallbladder, posteriorly to portal vein, gastroduodenal artery and common bile duct.
- Second part: descends in front of right kidney; receives bile duct and pancreatic ducts; related medially to head of pancreas.
- Third part: crosses in front of IVC and aorta at L3; related anteriorly to superior mesenteric vessels and root of mesentery.
- Fourth part: ascends to duodenojejunal flexure; related posteriorly to left psoas major, left sympathetic chain, left gonadal vessels and inferior mesenteric vein.

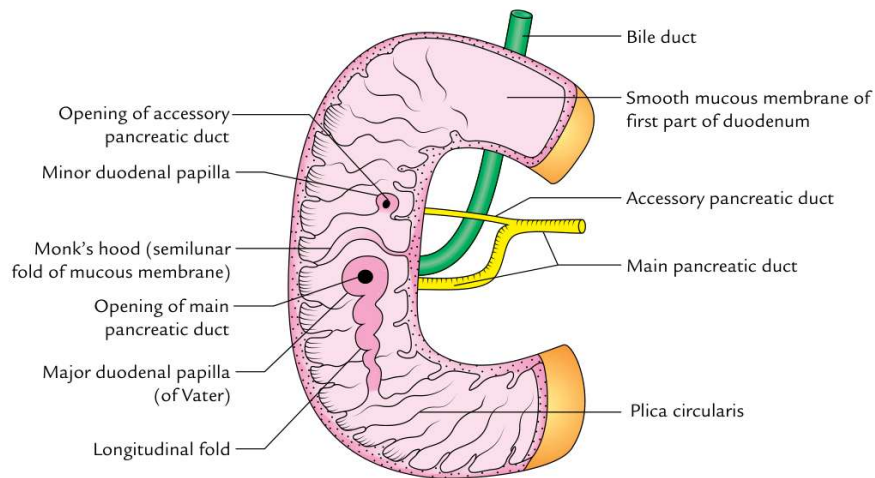


Fig. 9.7 Interior of the duodenum.

Fig. 9.7 - Interior of the duodenum.

The interior of the second part of the duodenum is exam-important. The major duodenal papilla receives the common hepatopancreatic duct, while the minor duodenal papilla receives the accessory pancreatic duct. Recognition of the major papilla is important during endoscopic retrograde cholangiopancreatography (ERCP).

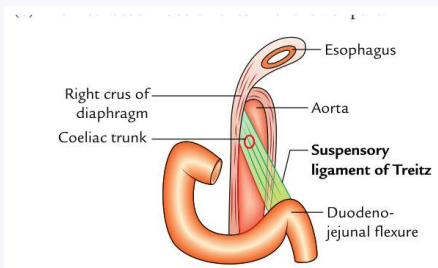


Fig. 9.8 Suspensory ligament of Treitz.

Fig. 9.8 - Suspensory ligament of Treitz.

Ligament of Treitz

The suspensory muscle of duodenum is a fibromuscular band extending from the right crus of diaphragm to the duodenojejunal flexure.

It fixes the duodenojejunal flexure and prevents it from being dragged down by the weight of small intestinal loops.

It is a useful radiological landmark in malrotation and may rarely kink the duodenojejunal flexure causing partial obstruction.

Duodenal arterial supply

- Upper half: coeliac trunk territory through superior pancreaticoduodenal artery.
- Lower half: superior mesenteric artery territory through inferior pancreaticoduodenal artery.
- Anterior and posterior pancreaticoduodenal arcades supply the duodenum and head of pancreas.

Duodenal clinical notes

- Duodenal ulcers commonly occur in the first part.
- Posterior ulcer may erode the gastroduodenal artery and cause severe bleeding.
- Perforation of an anterior duodenal ulcer may lead to peritonitis.
- Major duodenal papilla is important in biliary and pancreatic duct procedures.

JEJUNUM AND ILEUM

Extent and proportions

The small intestine proper extends from the duodenojejunal flexure to the ileocaecal junction. Its upper two-fifth is jejunum and lower three-fifth is ileum. There is no definite line of demarcation between them, so they are distinguished by progressive structural changes.

Jejunum - usual features

- Mostly lies in the upper left part of the infracolic compartment.
- Wider lumen, thicker wall and more vascular appearance.
- Circular folds are tall, large and closely set.
- Vasa recta are long and arterial arcades are fewer.

Ileum - usual features

- Mostly lies in the lower right part of the abdomen and pelvis.
- Narrower lumen, thinner wall and less vascular appearance.
- Circular folds are smaller and sparse distally.
- Peyer patches are larger, more numerous and placed along the antimesenteric border.

STRUCTURE OF THE WALL

Four layers from inside outward

- Mucosa: responsible for absorption; presents circular folds, villi, microvilli, intestinal glands and lymphatic follicles.
- Submucosa: loose areolar tissue containing blood vessels, lymph vessels and Meissner plexus.
- Muscle layer: outer longitudinal and inner circular smooth muscle with Auerbach plexus between them.
- Serosa: visceral peritoneum covering the mobile jejunum and ileum.

It is made up of outer longitudinal and inner circular smooth muscle. Auerbach's plexus of nerves is between these two layers.

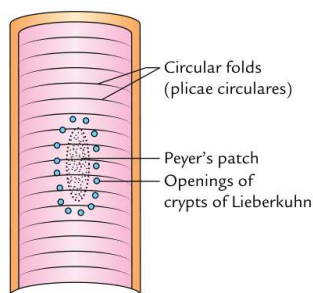


Fig. 10.2 Interior of ileum showing a Peyer's patch

Fig. 10.2 - Peyer patch in the ileum.

Mucosal features explained

Plicae circulares are permanent circular folds that increase absorptive surface area and are not obliterated by distension.

Villi are small finger-like projections on the mucosal surface; their epithelial cells bear microvilli, producing a very large absorptive surface.

Intestinal glands or crypts of Lieberkuhn open between villi and secrete mucus and enzymes.

Aggregated lymph follicles form Peyer patches, especially in the ileum. Their involvement explains typhoid and tubercular ulcers.

MESENTERY OF THE SMALL INTESTINE

transtubercular planes.

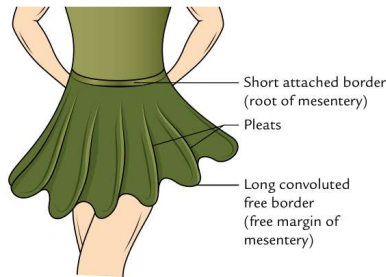


Fig. 10.3 Mesentery of the small intestine resembling a full skirt.

Fig. 10.3 - Mesentery resembles a full skirt.

Why the mesentery matters

The mesentery is a broad fan-shaped fold of peritoneum that suspends jejunum and ileum from the posterior abdominal wall.

It has a short attached border called the root and a long free intestinal border. Because the free border is much longer, the mesentery forms folds like pleats of a skirt.

It carries jejunal and ileal arteries, veins, lymphatics, lymph nodes, autonomic nerves and fat to the small intestine.

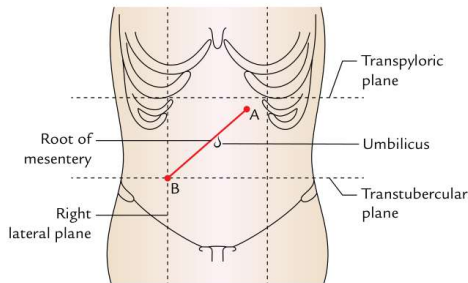


Fig. 10.4 Surface marking of the root of the mesentery: A, point representing duodenojejunal flexure; B, point representing ileocaecal junction.

Fig. 10.4 - Surface marking of root of mesentery.

Root of mesentery

The root is attached obliquely across the posterior abdominal wall from the duodenojejunal flexure to the ileocaecal junction.

The duodenojejunal flexure lies to the left of L2 vertebra. The ileocaecal junction lies at the right sacroiliac joint.

Surface marking: from a point 1 cm below the transpyloric plane and 2.5 cm left of the midline to the intersection of right lateral and transtubercular planes.

BLOOD SUPPLY, VENOUS DRAINAGE, LYMPH AND NERVES

Arterial supply of jejunum and ileum

The jejunum and ileum are supplied by jejunal and ileal branches of the superior mesenteric artery. These branches enter the mesentery and form arterial arcades. Straight vessels called vasa recta arise from the arcades and supply the intestinal wall.

- Jejunum: fewer arterial arcades and longer vasa recta.
- Ileum: more arterial arcades and shorter vasa recta.
- Terminal ileum: also receives supply from the ileocolic branch of the superior mesenteric artery.

Venous drainage

- Veins correspond to the arteries and drain into the superior mesenteric vein.
- Superior mesenteric vein joins splenic vein behind the neck of pancreas to form portal vein.

Lymphatic drainage

- Lymph vessels pass through mesenteric nodes.
- They then drain to superior mesenteric nodes around the origin of the superior mesenteric artery.

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- Portal drainage carries absorbed nutrients to the liver.
- Circular ulcers and strictures in tuberculosis relate to the circular course of intestinal lymphatics.

Nerve supply

The small intestine receives both sympathetic and parasympathetic fibres. Sympathetic fibres arise from T10-T11 spinal segments through splanchnic nerves and superior mesenteric plexus, while parasympathetic fibres come from the vagus through coeliac and superior mesenteric plexuses.

- Sympathetic fibres are motor to gut sphincters and inhibitory to peristalsis.
- Parasympathetic fibres stimulate peristalsis and inhibit sphincters.
- Pain from jejunum and ileum is commonly referred to the umbilical region.

JEJUNUM VS ILEUM - HIGH YIELD COMPARISON

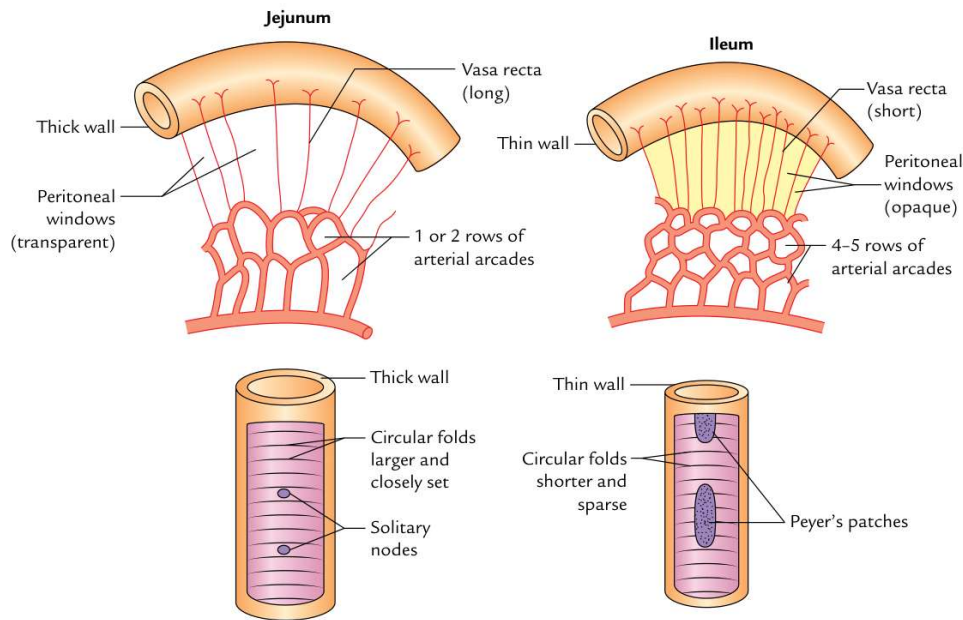


Fig. 10.5 Differences between the jejunum and ileum.

Fig. 10.5 - Differences between jejunum and ileum.

This is one of the most important operative and viva diagrams. Jejunum shows thicker wall, larger folds, fewer arcades and longer vasa recta. Ileum shows thinner wall, Peyer patches, more arcades, shorter vasa recta and more fat in the mesentery.

Table: Differences between jejunum and ileum

Feature	Jejunum	Ileum
Wall	Thicker and more vascular	Thinner and less vascular
Lumen	Wider; often empty	Narrower; often full
Circular folds	Long, large and closely set	Smaller, shorter and sparse distally

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Villi	More, larger, thicker and leaf-like	Fewer, shorter and finger-like
Peyer patches	Small, circular and fewer; mostly distal jejunum	Large, oval and numerous; maximum in distal ileum
Mesentery	Less fat; translucent peritoneal windows visible	More fat; peritoneal windows are absent
Arterial arcades	One or two rows	Four or five rows
Vasa recta	Long and relatively fewer	Short and more numerous

DEVELOPMENT AND MECKEL DIVERTICULUM

Development

Jejunum and ileum develop from the U-shaped midgut loop. The cephalic limb forms jejunum and upper ileum, while the caudal limb forms lower ileum. During embryonic life, the apex of the midgut loop communicates with the yolk sac through the vitello-intestinal duct, which normally disappears by the sixth week of intrauterine life.

Midgut rotation clue

- The developing midgut loop undergoes physiological herniation and rotation before returning to the abdominal cavity.
- Abnormal rotation can alter the position of duodenojejunal flexure and caecum.
- The ligament of Treitz and duodenojejunal flexure are therefore useful landmarks in imaging.

Vitello-intestinal duct remnants

- Complete persistence may form a fistula between ileum and umbilicus.
- Partial persistence may form Meckel diverticulum, fibrous band, sinus or cyst.
- These remnants can cause obstruction, bleeding, inflammation or acute abdomen.

Clinical importance of embryology

- Most congenital abnormalities of this region involve midgut rotation or persistence of the vitello-intestinal duct.
- A band from Meckel diverticulum to the umbilicus may trap intestinal loops and produce obstruction.
- Ectopic gastric mucosa inside Meckel diverticulum may secrete acid, causing ulceration and bleeding in the adjacent ileal mucosa.
- When acute appendicitis is suspected but the appendix is normal, the distal ileum should be examined for Meckel diverticulum.

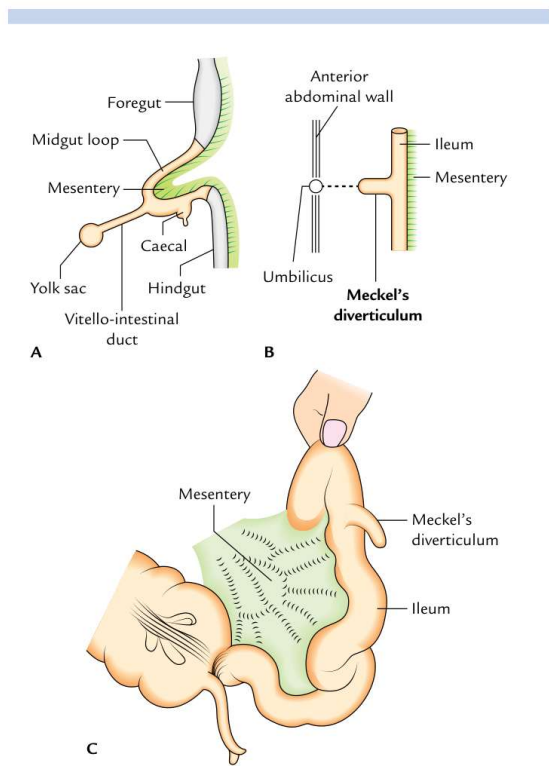


Fig. 10.6 Meckel's diverticulum: A, vitello-intestinal duct connecting midgut loop with yolk sac; B, Meckel's diverticulum (schematic representation); C, Meckel's diverticulum as seen during surgery.

Fig. 10.6 - Meckel diverticulum.

Meckel diverticulum: 2-rule memory

It is the persistent proximal part of the vitello-intestinal duct and is attached to the antimesenteric border of the ileum.

It occurs in about 2% of people, is about 2 inches long and is situated about 2 feet proximal to the ileocaecal junction.

It may contain ectopic gastric mucosa or pancreatic tissue, so it may ulcerate, bleed, perforate or mimic appendicitis.

During surgery for suspected appendicitis, if the appendix is normal, Meckel diverticulum should be looked for.

CLINICAL ANATOMY

Typhoid and tubercular ulcers

- Peyer patches may be felt through the antimesenteric border and may ulcerate in typhoid fever.
- Typhoid ulcers are usually longitudinal or oval and may perforate.
- Tubercular ulcers are circular; after healing they may cause fibrous narrowing and intestinal obstruction.

Mesenteric conditions

- Mesenteric lymphadenitis can mimic acute appendicitis.
- Mesenteric cyst presents as a cystic swelling in the umbilical region.
- Mesenteric cyst is more mobile at right angles to the line of attachment of the mesentery.

Obstruction and hernia

- Long mesentery allows loops of small intestine to descend into a hernial sac.
- Failure of the root of mesentery to fuse properly may form mesenteric parietal hernia of Waldeyer.
- Ligament of Treitz may rarely kink the duodenojejunal flexure.

Applied blood supply

- Duodenum is supplied by both coeliac and superior mesenteric systems.
- Jejunum and ileum are supplied mainly by superior mesenteric artery.
- Portal venous drainage explains why absorbed nutrients from intestine first reach the liver.

REVISION SHEET

Quick viva answers

Question	Answer
Shortest and widest part of small intestine	Duodenum
Most fixed part of small intestine	Duodenum
Mobile parts of small intestine	Jejunum and ileum
Upper two-fifth of small intestine proper	Jejunum
Lower three-fifth of small intestine proper	Ileum
Main artery of jejunum and ileum	Superior mesenteric artery
Main lymph nodes of jejunum and ileum	Mesenteric and superior mesenteric nodes
Important lymphoid tissue in ileum	Peyer patches
Surgical landmark for duodenojejunal flexure	Ligament of Treitz
Common rule for Meckel diverticulum	Rule of 2: 2%, 2 inches, 2 feet from ileocaecal junction

Last-minute memory hooks

- Jejunum = juicy: thicker wall, wider lumen, more vascular, large folds, long vasa recta.
- Ileum = immune: Peyer patches, more arcades, shorter vasa recta, more fat in mesentery.
- Duodenum = duct receiver: bile duct and pancreatic ducts open into its second part.
- Pain from jejunum/ileum is referred near the umbilicus because they are midgut derivatives.